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Baniyan Tree - *Ficus benghalensis*
Living Legacy of India

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நமது வனம்

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Editor's Desk

Artificial Intelligence (AI), as we all know, is the latest buzzword in whatever field you could assume. AI officially took birth at the Dartmouth Conference, at Dartmouth College located in Hanover, New Hampshire, USA, in 1956. This invention evolved over decades through the work of many scientists, mathematicians, and computer engineers. Notable contributors among them were Alan Turing, Marvin Minsky, Herbert Simon, and Allen Newell. Now, AI is gradually becoming the foundation for most technologies – medicine, defence, agriculture or name any activity or sector. It has not yet reached the threshold of inevitability, but is accelerating rapidly across most sectors.

We can safely assume the beginning of the concept of AI, was from this famous question "Can machines think?" by Alan Turing (1912–1954), inventor of “universal machine or the “Turing Machine” and the protagonist of foundation for theoretical computer science and AI with his “Turing Test” in 1950. Late, John McCarthy (1927–2011) coined the term "Artificial Intelligence" in 1955 and organized the first AI conference at Dartmouth College in 1956. He created the Lisp programming language, which became central to early AI research. And hence he is aptly called the Father of Artificial Intelligence. While, the Co-founder the MIT AI Lab. Marvin Minsky (1927–2016) promoted the idea that the mind could be replicated with machines. Allen Newell (1927 -1992) & the Nobel Prize winner in Economics (1978) Herbert A. Simon (1916-2001), both demonstrated that machines could perform reasoning tasks traditionally thought to require human intelligence and also created first AI programs like Logic Theorist (1955) and General Problem Solver (GPS).

Now, the question is can we, the foresters, use AI in our profession?

Answer, is an emphatic “Yes.” Countries like United States, Australia, India and Canada are already using AI in Forest fire detection and prediction. Forest Survey of India uses AI to monitor forest fires via MODIS and SNPP satellite data. Illegal Logging Detection is being carried out with AI's help by Brazil and Indonesia. Canada, Finland and Germany are already doing Forest Inventory & Biodiversity Monitoring through AI analysis of drone/LiDAR imagery to estimate tree species, height, biomass and detect forest health and disease outbreaks. Finland's Metsähallitus, uses AI + drone imagery for sustainable forest planning and biodiversity tracking. Using Machine learning models,

countries like Kenya, India, U.S., South Africa, classify species from millions of camera trap images. Examples are plenty.

The Tamil Nadu Forest Department, in a ground breaking achievement using AI, has developed an outstanding AI based Warning System, which provides real-time alerts to loco pilots and patrol teams on the presence of elephants on the railway tracks. This country's first Artificial Intelligence (AI) and Machine Learning-Enabled Surveillance System, designed to prevent elephant deaths on railway tracks by speeding trains, has proven to be successful along the busy interstate route between Palakkad in Kerala and Coimbatore in Tamil Nadu. Installed by the Coimbatore Forest Division, since its inauguration in February 2024, the AI Surveillance System has effectively safeguarded the traditional migratory path in the Walayar-Madukkarai Forest Range. Notably, after installation of this early warning system, no wild elephants have been killed on these two railway tracks, marking a significant success for elephants' safety.

Another potential use of AI in wildlife management is that, most Forest Departments have huge recorded data generated by Anti-poaching patrols. These data are now, a treasure trove. It will be easy to analyse these observations and generate usable data using AI, for monitoring animal movements and the areas where they are concentrated. Based on these findings carrying capacity of the locations can be determined more scientifically.

AI deployment may pose some of challenges like: a) access to high-quality satellite or drone captured data, which can be expensive or restricted; b) mis-interpreting some events or noise, which may act as false positives; c) labour intensive, time consuming and costly ground truthing and combining local expertise, to validate the findings.

AI doesn't replace human element in forest protection — it can help in amplifying their capacity. With the right data and partnerships, AI can drastically and qualitatively improve how we monitor, protect, and restore wildlife and forests.

To conclude, we look for the day when the Tamil Nadu Forest Department has a dedicated AI Unit to capitalise on this AI technology and contribute for an improved and scientific forestry practices in our State of Tamil Nadu and on the whole in our Country.

Let us hope things do happen positively.



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Former Additional Principal
Chief Conservator of Forests

REMINISCENCES OF A FORESTER - SOME SNIPPETS (Episode 4)

Brief tenure in line

Since Cuddalore division wasn't supporting economically valuable forests, most of my time was devoted to the pioneering experiment of mangrove restoration in Pichavaram, other than undertaking Eucalyptus plantations in the plain forests of Kallakurichi and Ulundurpet Ranges. As the country celebrated its Independence Day in 1988, a flagship project in the name of 'Independence Day Ecology Farm' was launched in all districts and the same in South Arcot was inaugurated in Muttakadu RF of Gingee Range. All planting and soil moisture conservation works under this signature project with focus on signage and publicity boards were in full swing.

While so, I received a government order posting me as District Forest Officer of Vellore Forest Division of North Arcot district in October 1988. I felt elated on three counts. I am posted to territorial task for the third time in a row; I was moving to one of the most prestigious and sensitive forest divisions in the State; It was as well the largest division of Tamil Nadu with over 1,84,000 ha of Reserved Forests and the highest number of around 360+ forest staff. As I landed in Vellore and ascended the steps of the District Forest Office within the ramparts of the historical Vellore Fort, I was recollecting the days, when I used to go to the same building as the probationary Assistant Conservator of Forests and Polur Range trainee only three years back. Familiarity of the landscape was a comforting factor that gave me some confidence about my 'going - to be innings' here.

Sandal, the revenue spinner

Major portion of forests of Vellore division fell in the Javadhis, a hill range in the Eastern Ghats. Assorted vegetation clothed the outer slopes as well the plateau in the hill that supported distinct forest types. While the lower slopes were dominated by the southern tropical thorn forests and scrub, dry deciduous and riverine vegetation occupied the upper slopes and plateau. In addition, certain isolated blocks of forests in the North, North-Eastern part of the district were constituted as Arcot Range that mostly contained artificially planted stands of red sanders (*Pterocarpus santalinus*) and Achan (*Harwickia binata*). Sandal (*Santalum album*) formed the principal crop of commercial importance with sizeable population of the species occurring in the non-forest lands in the 100 and odd forest settlements of Malayalee tribe in Javadhi hills.

Prescription followed in the working of sandal was to extract only the dead, diseased and wind-fallen trees in a felling cycle of three years. Cutting of green sandal trees of whatever girth or height wasn't permitted as per Working Plan. Rigorous sandal felling rules were in force that required the checking of enumerated dead trees in the annual sandal coupes, verification of extracted trees with

reference to Extraction and Conversion (E&C) form in the forest depots, checking of the pits after extraction by senior officers up to the level of Conservator of Forests. I am not sure as to how many of the present-day forest officers of the State are conversant with the E&C forms with no harvestable sandal standing crop in the forests. The local tribe were engaged in the entire range of activities connected with sandal harvest and transport to the depots under the supervision of field officials. Annually few thousand trees used to be extracted and transported to the Tirupattur sandalwood depot that brought substantial revenue to the division. Vellore was a revenue surplus division then. All the sandal bearing ranges were provided with jeeps to conduct intensive patrolling and offer protection to sandal wealth.

When I joined the division, there wasn't noticeable threat in form of organized sandalwood smuggling other than sporadic cases of illicit cutting, carriage on headload to the foot hills by the Malayalee, which then got transported by two wheelers, cars and other light motor vehicles by people from the plains. Sandal theft wasn't that rampant so much so that hundreds of harvested trees were left in the forest depots, mostly close to motorable forest tracks for months together for due checking by senior officers before they were transported to Tirupattur. The beat staff were supplemented with a set of sandal tending watchers, who were tasked with climber cutting in sandal trees, bush sowing of seeds besides assisting the field formation in sandal protection and harvest related works. The division had established a network of about 20 forest check posts at vulnerable exit points from the forests, where watchers on muster roll were on duty. During our monthly Conservator's review meeting, I was hearing from my counter part of Tirupattur Division that intense heat of organized smuggling was already felt in his jurisdiction.

Another spell of pioneering work

Many plantation works were implemented in the Division and the chief species were Eucalyptus hybrid and bamboos under Pulpwood scheme. For the first time in the country, a pilot project of Aerial Seeding for ecological restoration of degraded forests was launched in Vellore Division during 1988-89 under the aegis of National Wasteland Development Board (NWDB). About 1,000 ha of degraded forest slope and plateau in Pallikonda RF in Vellore Range was selected for treatment under the project that included raking of soil at intervals along the contour and sowing of pelletized seeds from Vayudoot aircraft in the north east monsoon season. Pelletization of pretreated seeds of over a dozen assorted species suited to the site was undertaken by the Genetics Division of Research wing, Coimbatore. Rest of the works were carried out by the officers of Vellore Division. As this was the first ever project of unique kind for Tamil Nadu, it drew lot of attention from the higher ups including field visits of the Vice Chairman of NWDB, Forest Secretary and Principal Chief Conservator of Forests. I vividly recollect the day when Shri Duleep Mathai the Vice Chairman, NWDB ascended the steep slope of Pallikonda RF to reach its plateau, accompanied by me, the Deputy Conservator of Forests (Genetics), Conservators of Forests, Vellore and Research Circles and the

Principal Chief Conservator of Forests to look at the work for himself. This unique massive eco-restoration programme was extended to about 5,000 ha in RFs of Ambur and Gudiyatham ranges during 1989-90, besides few other divisions across the state. All tasks multiplied five times!

Having completed the sowing that followed site preparation, onus for developing a monitoring protocol for assessing the outcome of aerial sowing and devising a 'special format' plantation journal for documenting the species wise germination, establishment and survival of seedlings in the sample plots laid in the treatment area fell on me. This task took considerable time from my work routine during 1988-1990, as besides taking up personal inspections I had to accompany senior officers who made repeat visits. The result was of mixed nature with few of the species like neem, custard apple, elandai, kodaivel, parambai displaying good performance. Later on, when I visited the area as the Regional Chief Conservator of Forests of Chennai Region during 2010 and 2011, I enquired with interest about the status of regeneration in those areas, which were indicated to be reassuring. I had a glimpse of the outer slopes of Pallikonda and Sanakuppam RFs facing the Vellore-Bangalore NH, which by then were bearing well stocked vegetation with moderately dense canopy cover.

Dynamics of sandal smuggling

Alongside implementation of afforestation works, I was concentrating on protection of sandal wealth of the division, which by the end of 1989 was showing signs of escalation. This was reflected from the quantum of seizures, number and category of vehicles caught in transit with illicit sandalwood, and the number of accused apprehended that registered an upward trend as compared to the previous year. I encouraged the Forest Range Officers to set up secret information system to obtain critical and timely information about illicit cutting of sandal trees and smuggling in their jurisdiction. Of course, Secret Service Funds came in handy. One such input from a local informant pointed to the arrival of a huge consignment of sandal logs to a plateau above the Pallakollai village in Santhavalas Range (from where many offences were booked earlier). Known as *karisuttanmandai* (a flat terrain within Amirthi RF where charcoal making was carried out in the past), this place was located 6-7 km uphill from the Amirthi-Jamunamarathur forest road above the Amirthi Zoo. I led a special party to the spot on foot only to encounter a score of head loaders from both the tribal settlements and the village at the foot hill. We were able to apprehend a few persons, as rest of them fled the scene on sighting us. There were scattered heaps of sandal logs, which had to be transported on headload by the members of the combing team down to the road. When weighed, the seizure accounted for about 7,000 kg of rough cleaned sandalwood. Such was the scale of operation!

Deliberations in the state level meetings in early 1990s pointed to certain policy and legal inadequacies as far as sandal protection was concerned. At the state level, a couple of factors contributed to the increasing trend of illegal felling. The trees standing on patta lands and in the tribal

settlements were the property of Government in general. For those dead trees extracted from settlement areas assigned to individuals prior to 1907 and 1928 in the Javadhi, the pattadars were entitled for about 47.5 per cent revenue sharing. This compensation never came in time as a result of which the pattadar had not shown any interest in the protection of patta sandal tree. For the other pattadar who aren't entitled for any revenue sharing, there was no incentive to protect the tree. They always saw a sandal tree standing on his land as a burden or liability. So, they either connived with the illegal cutters or turned a blind eye for the fear of retribution. In addition, the rates for felling, conversion, transport of sandalwood from forest coupes to the depots were found to be grossly inadequate considering the intensity and quantum of labour involved. The tribal tended to leave many large-sized dead sandal trees in the deep interior of the forests for want of sufficient labour rates. Such trees fell victim in the hands of the very tribal who left them unmarked for quick return.

At country level there was no uniform legislation for possession, transport, value addition and trade of sandal wood and its products across states. At that time, possession of sandalwood was not an offence in some of the states and there were also no rigid transit rules for the species. In some States, no license was required from the State Government in the Forest Department for the purpose of starting or renewing sandal oil distilleries. Export of Sandalwood was permitted in the form of chips whose average weight is 50 grams and less. All these contributed to the seamless mixing of illegally sourced sandal with the legitimately procured wood from auction at government depots.

Intensive Combing Operations

When I felt that the scale of clandestine movement of illicit wood was rising, I introduced a system of 'Intensive Combing of forests' by special combing teams formed with the jurisdictional staff and special duty staff. These teams moved in tandem to perambulate vulnerable forest tracts and the abutting tribal hamlets, besides keeping vigil over the night time movement of vehicles along forest margins. This led to interception of many sandal-laden vehicles on movement close to the forest boundary. On many occasions, smugglers adopted devious methods to conceal the wood consignment. It was a truck loaded with either onion, potato or cabbage sacks on top with sandal logs neatly concealed underneath. Sometimes, it was noticed that few bottom line temporary watchers were found to be in collusion with the local smugglers, which on detection were dealt with appropriately by including such persons as accused in the offence. An incident that occurred on the wee hours of the new year day of 1990, when one Forest Range Officer blocked an Ambassador car with the spiked wooden plate (which was commonly deployed to halt suspicious vehicles) on a village road. To his surprise, he noticed that the vehicle's back seat and dicky was fully loaded with sandal billets. On enquiry, he found that the person on wheel and one sitting nearby, both in lungi, were constables from Forest Cell CID. They were promptly apprehended. The incident was taken to the notice of higher-ups in the department. It is another matter that the Forest Cell CID established in 1983 to investigate offences involving interstate ramifications was wound up subsequently in the late 1990s.

Administrative and Management interventions

When it was realized that the schedule of rates for felling, billeting, extraction of roots down to diameter of 50 paise coin from pits and head loading to forest depot was one among the reasons for leaving large size dead trees unmarked exposing them to theft, the Department commissioned a time-motion study that resulted in reasonable upward revision of Schedule of Rates in 1990. Impact was visible soon and many hundreds of huge dead trees were marked and harvested in most of the Sandal divisions. The legal backup by way of including the forest offenders during 1986 within the ambit of Goondas Act 1982 had its salutary effect in containing the smuggling menace to some extent. The first ever order of detention of a habitual sandalwood offender under the Goondas Act in Tamil Nadu was passed by the District Collector, Vellore upon the recommendation of Forest Range Officer, Arni Pulpwood Range with headquarters at Polur during my tenure in 1990. Thereafter many offenders were brought to book under the Goondas Act. A special scheme of establishing 'sandal nucleus estates' to promote regeneration of the species was also under implementation in five divisions, one of which was Vellore division.

Epilogue

Yet, it was running around for the forest field staff including night raids under the supervision of the divisional Assistant Conservator of Forests and under the control of the District Forest Officer to mitigate the surging smuggling menace. On most of the festival eves, we used to be on the roads at nights only to return home at dawn. This continued till my departure from the division in May 1991. Around middle of 1990, the Conservator of Vellore circle requested for the status of sandal in various divisions under his jurisdiction. I remember to have submitted a detailed narration about the happenings in this sphere and summed up stating that the future of sandal was at stake and mature stock of trees would be wiped out of its' traditional strongholds by the turn of the century, unless multipronged strategy involving policy, legal and administrative measures are put together in full earnest. My premonition proved right, as all of us learnt soon thereafter.

To be continued...



Dr. T. Sekar, IFS., (R)
Former Principal Chief Conservator
of Forests

THE ETERNAL CANOPY:SACRED SYMBOL AND LIVING LEGACY OF INDIA

Towering with grace, spreading with wisdom, and deeply rooted in time, the Banyan Tree (*Ficus benghalensis*) is not just a botanical marvel—it is a living symbol of India's cultural, spiritual, and ecological heritage. With aerial roots descending like veils from its vast limbs, the banyan becomes not one tree, but an entire grove—a timeless presence sheltering generations of beings beneath its canopy. This majestic tree has watched civilizations rise and fall, sages meditate, animals seek refuge, and stories unfold beneath its shade. As we journey into the legacy of the banyan, we uncover not just the story of a tree, but of a living entity entwined with the soul of the land.

Historical Roots: The Banyan in the Timeline of India

The banyan has been revered in Indian history since the Vedic era. Its name originates from the word “banias”, referring to the merchants who often conducted trade beneath its cool shade. These natural gathering spots became village centers, market spaces, judicial forums, and venues for philosophical discourse. In fact, the banyan often stood in the heart of villages and temple towns as a natural town hall—its shade fostering dialogue, justice, and community bonding.

One of the oldest living banyans, the Great Banyan Tree at Acharya Jagadish Chandra Bose Botanical Garden in Kolkata, is estimated to be over 250 years old and covers more than 3.5 acres—symbolizing its ancient legacy and the idea of perpetual regeneration.



Plate 1- Roots that breathe, fruits that nourish, leaves that shelter - the banyan's living embrace.





Fig.1- The majestic Banyan of the Theosophical Society, Chennai - a living witness to history, sheltering visionaries from Gandhi and Tagore to Annie Besant and Sarojini Naidu beneath its timeless canopy.



Fig.2- The Great Banyan of Kolkata — a 250-year-old living monument, weathering storms and centuries, its countless roots weaving a forest where time itself seems to rest in shade. A postal stamp was issued by the Indian Postal Department to commemorate this Banyan Tree in 1987.



Fig. 3- Nature's cathedral — the world's largest banyan tree, Thimmamma Marrimanu, standing strong for over five centuries.

Ecological Majesty: The Banyan as a Keystone Species

Banyan trees are among India's most iconic natural landmarks, valued for their ecological, cultural, and spiritual importance. Three of the largest and most historic banyans showcase both nature's resilience and India's deep-rooted traditions.

1. Thimmamma Marrimanu, Anantapur district, Andhra Pradesh (c. 550 years old)

Recognized by the Guinness World Records as the world's largest banyan, this sacred tree covers 19,107 m²—almost four football fields. It is a pilgrimage site where devotees believe blessings of fertility and family well-being are bestowed.

2. Adyar Banyan Tree, Theosophical Society, Chennai (c. 450 years old)

Once India's largest banyan with a spread of nearly 40,000 m², it partially collapsed in the 1990s due to natural decay and cyclones. Still, it continues to stand as a symbol of spiritual gatherings and philosophical discourse.

3. The Great Banyan Tree, Acharya Jagadish Chandra Bose Botanical Garden, Kolkata (c. 250 years old)

With a canopy of 18,918 m², this tree survived the loss of its main trunk in 1925. Its forest-like structure of aerial roots continues to inspire scientists and visitors alike.

These monumental banyans, each centuries old, reflect India's living heritage—where ecology, spirituality, and culture are interwoven under the vast shade of nature's wonders.

Ecologically, the banyan is a cornerstone of tropical biodiversity. It serves as a keystone species in Indian ecosystems, supporting hundreds of organisms through its massive canopy and fig-bearing branches. Birds such as parrots, mynas, and bulbuls feast on its fruit; bats, bees, butterflies, and beetles seek its nectar and shade. Its hollow branches and aerial roots are nesting grounds for owls, civets, and snakes.

As a fig species, it is part of the *Ficus* genus, which plays a unique ecological role through “obligate mutualism” with specialized wasp pollinators. This intricate relationship sustains not only the tree's propagation but also ensures the survival of many other plant and animal species that depend on its existence.

Moreover, the banyan's roots bind soil, prevent erosion, and improve microclimates. Its dense canopy acts as a carbon sink, making it an important contributor to climate resilience in urban and rural landscapes alike.

Spiritual Sanctity: A Living Temple

To millions across India and Southeast Asia, the banyan is not merely a tree—it is a sacred living being. In Hinduism, it is considered immortal and associated with Lord Vishnu, Lord Shiva, and the goddess Lakshmi. The tree represents the Trimurti—with the roots symbolizing Brahma (creation), the trunk Vishnu (sustenance), and the branches Shiva (destruction and regeneration).

The Vat Savitri Vrat is a prominent festival where married women worship the banyan tree, tying sacred threads around it and praying for the longevity of their husbands—honoring the tree's symbolism of enduring love and life.

In Jainism, it is believed that the Adinath (the first Tirthankar) and Padmaprabha (6th

Tirthankar), attained Kevala Jnana (omniscience) under a banyan tree. Likewise, in Buddhism, it is considered a cousin of the Bodhi tree under which Siddhartha Gautama attained enlightenment.

The tree is also seen as a cosmic symbol—the “Ashvattha” in the Bhagavad Gita, an upside-down tree with roots in the heavens and branches on earth, representing the link between the material and spiritual realms.

Cultural Embrace: The Tree in Art, Literature, and Life

The banyan has inspired poets, painters, sages, and environmentalists. Rabindranath Tagore described it as “the silent witness of the cycles of life.” Its imagery appears in ancient scriptures, folk tales, temple architecture, and modern art alike. Villagers believe that ancestral spirits reside in the tree; thus, they avoid cutting it, preserving its legacy through oral tradition and spiritual caution.

In traditional Indian paintings and temple carvings, the banyan is depicted as a seat of sages, where disciples gather for spiritual teaching, echoing the ancient Gurukul system. The banyan thus becomes not just a tree but a university of wisdom under the sky.

Conservation and Challenges: Saving the Silent Sentinel

Despite its revered status, the banyan today faces threats from urban expansion, infrastructure projects, pollution, and neglect. Old banyan trees are being felled to widen roads or make way for modern construction. Invasive concrete and tar at its base prevent the tree's roots from breathing, weakening its foundation.

Fortunately, conservation efforts are gaining momentum. Institutions like the Botanical

Survey of India, NGOs, and local communities are now identifying and protecting heritage trees. Movements like “Sacred Grove Conservation”, urban biodiversity parks, and tree-mapping programs are helping revive awareness.

The banyan's inclusion in India's national emblem—the symbol of resilience, unity, and shelter—reminds us of our collective responsibility to safeguard this natural heritage.

The Banyan Tree as a Timeless Teacher

In an age where concrete replaces canopies and digital screens overshadow deep-rooted **traditions**; the banyan stands as a timeless teacher. It teaches patience in growth, strength in stillness, generosity in shelter, and interconnectedness in existence. A witness to births and funerals, weddings and wars, meditations and markets—the banyan is not merely a tree. It is a legacy, a legend, a living bridge between nature and culture.

Let us honor this sacred sentinel not just in ritual but in action—by protecting, planting, and passing on its story to future generations. The eternal canopy must remain—not just in forests and parks, but in our collective consciousness.



Plate-2 The sacred Banyan — worshiped in rituals, its sticks (ஆலமரக் குச்சி) used in homam, pooja, and even as natural toothbrushes — a true keystone species sustaining life and tradition.





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A FORESTER'S ORDEAL: A PLANTATION PROTECTION RAID



Tank Foreshore Babul (*Acacia nilotica*) plantation

In the early 1990s, I was transferred within the Social Forestry Division from Kovilpatti to Kuruvikulam Range. I departed one morning for my new station. My first day in Sankarankoil brought me to town around 1 PM. I quickly secured a room in Assembly Lodge, opposite the old bus stand, settled in, and then sought out a nearby restaurant for lunch. Just as I prepared to eat, I witnessed a man at the adjacent table being struck on the head with a chair by an assailant, blood splattering across the food. This violent, yet strangely captivating, incident immediately marked a traumatic and thrilling start to my time in Sankarankoil.

A year later, my Range was merged with the Sankarankoil Social Forestry Range, a new formation developed from the former Farm Forestry Range. This expanded Range now boasted numerous tank foreshore babul plantations, ripe for

harvesting. Our team headed by Mr. R. Rajasekaran as Range Officer who was very friendly, young, energetic and unified force also, and with eight Foresters including me. We foresters all were young, very positive and cooperative with Range Officer. We also supervised over a hundred plot watchers and village social workers.

During that period, Sankarankoil was, and still is, considered a highly sensitive area. Those days illegal arrack distillation was rampant, with many families from specific villages within the Sankarankoil taluk were deeply involved in the unlawful production. More than 150 to 200 people participated daily in debarking babul trees and each collecting bark from 50 to 100 trees. They transported this bark by head load and sold it to arrack distillers, as it was a primary raw material for illicit distillation.

Debarking babul trees can severely damage their xylem and phloem, ultimately causing the trees to wilt and die. When this occurred, local residents would harvest the dead trees for firewood, resulting in significant financial losses for the government and severe disciplinary action for the Foresters incharge of the plantation. To combat this, our team initiated a regimen of unpredictable patrols across every plantation. At that time, our DFO, Mr. R.G.S. Natarajan, with the repeated insistence of the Ranger, provided us with a new publicity van specifically for our protection efforts.

One day, our Ranger led all the Foresters, including me, on a joint protection raid, leaving behind the plot watchers and social workers. The Ranger Officer initially instructed Mr. Mariappan, a 55-year-old Forester, to remain at the Range Office due to his age. However, as we were about to leave, Mr. Mariappan approached me and secretly expressed his desire to join us. I then asked the Range Officer to permit him to come along, and we all proceeded to the Kandigaiperi Plantation in the Sankarankoil section.

We got to the plantation around noon and got out of our van. We walked about 200 meters into the thick trees. We heard talking in the distance, so we ducked down and looked. We saw the legs of over a hundred people working, stripping bark from trees. Two of our colleagues had wandered off on their own, ignoring the range officer's warnings and seemingly unaware of any potential trouble. The six of us, Foresters, all between 35 and 45 years old, along with the Range Officer, silently watched the scene unfold. Suddenly, the workers spotted us. They started yelling, "Only seven of them! We'll get all the officers!" Then they started throwing rocks and moving around us to trap us. As they came closer, we slowly walked backward. When they started moving faster, we ran as fast as we could back to our van. The van was running, ready to

leave, but we couldn't go without our two friends. We waited anxiously, but they didn't show up for a long time. Eventually, the group stopped chasing us, and their voices got quieter.

Time ticked by slowly. To our relief, Mr. S. Mariappan, the Forester, eventually emerged from the plantation, murmuring, "Was that fair? To abandon me alone in there?" We quickly calmed him down, and he began to recount what had happened inside the babul plantation. He'd been surrounded by the mob, and two men had tried to attack him with folded bunches of babul bark. Mariappan, thinking quickly, shouted for them to stop, reminding them that his daughter was married in their village and naming his son-in-law. This unexpected connection defused the situation; the mob released him. Mariappan then shrewdly told the group that higher-ranking officers were on their way for an inspection and would identify everyone involved in the debarking. His words had the desired effect: the mob, now pacified, gathered their collected babul bark and returned to their village. A courageous man happened to be in the office and then completely changed the situation.

We still had one missing forester, Mr. V. Jeyachandran, who was in charge of that particular plantation. An hour passed, and he still hadn't appeared. We grew increasingly anxious; preparing to search for him, worried the mob might have assaulted him. Just as our concern peaked, our Range Officer spotted him. Mr. V. Jeyachandran was making his way towards us from a nearby Ariyur hillside, and the Range Officer quickly calmed our fears. He explained that as he entered the plantation, he'd seen the crowd in the distance and immediately fled to the top of the nearest hill to avoid them. From that vantage point, he observed the mob's movements until they all returned to their village. Once he was sure the coast was clear, he came to find

us. With everyone accounted for and safe, we left for our office. Unfortunately, incidents like these were not uncommon in our area.

After our harrowing experiences, we made a crucial decision: all future protection raids would only be conducted with sufficient personnel, including plot watchers and social workers. This new approach, united under the leadership of Mr. R. Rajasekaran, our Forest Range Officer, allowed us to successfully protect entire plantations without a single tree being felled.

We raised wonderful plantations also in spite of conducting many raids; if we mentioned them all, it would become more voluminous. One time, the Range Officer, eight Foresters, and over fifty plot watchers rode in a lorry and hid in the Ariyur hillock by 5:00 AM. Within two hours, more than fifty head loaders were seen walking toward the hill. But because the land was so rough, almost everyone escaped when we got close. Few people, including one woman, were caught. We let the people we caught go after giving them a serious warning. This warning was given with the Panchayat President and other village elders present, and only after they gave us a written promise. We faced significant obstacles and threats throughout this period. It was solely due to the unwavering integrity and dedication of the plot watchers and social workers, who numbered over 130, that both we and the plantations were protected. During that time, an astonishing 1,500 to 2,000 hectares of A category babul plantations were sold in the Sankarankoil Range.

During that same turbulent period, the neighbouring Puliangudi Range faced even greater challenges. Four entire plantations were completely cut down and carted off by villagers, with the police unable to intervene due to significant law and order concerns. Months later, a Forester in the Puliangudi Range was brutally attacked with a weapon by a mob while trying to prevent tree felling in a social forestry plantation, tragically costing him an eye. Only after this severe assault did the police finally file a case.

Thirteen years later, as the Range Officer in that very same Puliangudi Social Forestry Range, I had the opportunity to revisit one of those previously plundered coupes. Recognizing its fertile soil, I selected it for the Greening Community Land Scheme, a project spearheaded by our then-DFO, Mr. K.R. Varadarajan, IFS., With the crucial cooperation of the local community, this new plantation thrived. It was diligently protected and, a decade later, successfully harvested by my colleagues. This experience truly showed me how time can change people and perspectives.

These were some of the most dramatic and impactful experiences we faced while working in the Social Forestry Range.



G. Sivagurunathan

Former Assistant Conservator
of Forest

TACTFULLY MANAGED BY A TOUGH FOREST OFFICER

In 1981, when I was serving in the Kalakad Range of Tirunelveli district, it was part of the Kalakad Wildlife Sanctuary (KWLS), originally established in 1962. Later, in 1988, both the Kalakad (251 km²) and Mundanthurai (567 km²) Wildlife Sanctuaries were merged to form the Kalakad–Mundanthurai Tiger Reserve (KMTR).

At that time, my batchmate Mr. Govindasamy was serving as the Forest Range Officer (FRO) of Ambasamudram Range, popularly known as Ambai Range, within the Mundanthurai Wildlife Sanctuary (MWLS). Both sanctuaries then came under the control of the Conservator of Forests (CF), Tirunelveli Circle.

Late, Mr.C.R.Thirumurthy, the was the then Wildlife Warden (WLW) of KWLS, which was under the Tirunelveli Forest Circle, headed by Late Mr. S. Sivasailam, Conservator of Forests, one of the toughest officers of those times. I am can still vvividly remember the tension and apprehension, when both WLW and CF came for field inspection. along the Manimuthar–Manjulai estate road.

The Manjulai estate 8,373.57 acres, part of Singampatti zamin forests, was leased to M/s Bombay Burmah Trading Corporation (BBTC), by the Singampatti Zamin on 12 February 1929 for 99 years to raise tea, coffee, and cardamom plantations, at a meagre rent of Rs. 1.70 per acre. The BBTC lease will expire on 11 February 2028. In contrast, the Government's Tamil Nadu Tea Plantation Corporation (TANTEA) pays an annual lease rent of Rs. 2,500 per hectare.

After the enactment of the Zamindari Abolition Act, the Singampatti forests was taken

over by the Tamil Nadu Forest Department in 1956, thereby becoming the lessor by stepping in to the shoes of Zamin, by virtue of take over. Following a prolonged legal battle spanning four decades, the Government of Tamil Nadu declared the entire 23,000 hectares of the Singampatti forest block as Reserve Forest (RF) with effect from 19 September 2018.

When BBTC approached the Madras High Court seeking patta and continued possession of the leased lands, the Court permitted the company to retain control only over the leased area until February 2028.

The Singampatti forest block is regarded as the core area of the KMTR, being home to rare and endangered species such as the Lion-tailed Macaque, Nilgiri Langur, Elephant, Gaur, Sambar, Barking Deer, Mouse Deer, Nilgiri Marten, Tiger, and Panther. This area also forms an ancient migratory corridor for elephants.

As the Ambai Range included the Manjulai area within its jurisdiction, Mr. Govindasamy also joined our inspection. The WLW of MWLS, however, could not attend the field camp. The jurisdiction of the Kalakad Range extended up to Kakkachi along the same road, with the eastern side of the road forming part of KWLS and the western side coming under Ambai Range.

As usual, I carried the map of the Kalakad Range, and the boundary details were discussed in detail with the CF.. Enroute near Marappalam, a British-era bridge, we were fortunate to spot a solitary Lion-tailed Macaque in the canopy. The sighting of this rare and elusive primate delighted the CF immensely.

Lion-tailed Macaque (*Macaca silenus*)



The Lion-tailed macaque is an endemic species of the Western Ghats of India. Its body is covered with dense black fur, while a striking silver-white mane encircles its face from the cheeks down to the chin. Both males and females share this distinctive appearance. The tail, ending in a black tuft of fur resembling a lion's tail, gives the species its common name. In Tamil, it is known as 'Singa Vaal Kurangu'.

These macaques are arboreal (tree-dwelling) and diurnal (active during the day). They usually live in social groups of 12 to 20 individuals, sometimes even more. The male's call is deep and resonant, and is said to resemble the human voice.

The Lion-tailed macaque is omnivorous, feeding on fruits, leaves, insects, small animals, and other forest produce. It has special cheek pouches to store food while foraging, an adaptation that helps it avoid competition and predators.

Currently, the Lion -Tailed Macaque listed as 'Endangered' on the IUCN Red List. due to loss and fragmentation of its and poaching, Conservation of its rainforest habitat is crucial to ensure the survival of this unique and charismatic primate.

As we continued on our return inspection, after a short break for refreshments, CF noticed a heap of freshly cut branches from various tree species lying by the roadside.

Known for his short temper and high concern for forest protection, his temper flared to the boiling point!. He, almost charged towards the FRO, Ambai, in whose jurisdiction the area fell, and reprimanded him severely. We remained frozen and silent for a while.

After a few minutes, I quietly enquired the Forest Guard (FG) of Ambai Range about the Forester in charge of the concerned section. In a low voice, he replied that the Forester and the Forest Watcher (FW) had gone to a nearby estate to apprehend the accused. He explained that the branches had been cut and brought from that estate, but prior permission from the Wildlife Warden (WLW) was required before removing shade trees within the tea estate.

I discreetly conveyed this information to the WLW, who in turn informed the CF that the Forester and FW were already on the job, tracing the culprits. After much persuasion, the WLW managed to calm the CF, who finally instructed the FRO, Ambai, to book an offence for pruning shade trees without prior permission.

We then resumed our journey and returned to the Manimuthar PWD Rest House for lunch. It was the first time any of us had encountered the CF in person, and the experience of dealing with such a moody and authoritative officer was both tense and memorable.

That evening, on our way back from Manimuthar, the CF was taken to inspect the tank foreshore babul plantation at Veetrirunthan Kulam, near Veeravanallur. The plantation extended over

about 60 hectares, excluding a submergible area of roughly 40 hectares. It was in excellent condition, well maintained, and thriving. The field staff had just begun planting boundary stones, only a few had been erected, while the rest lay scattered on the ground. On seeing this, the CF instructed us to ensure that the planting of boundary pillars was completed promptly, warning that any delay could lead to complications in future.

While crossing Cheranmahadevi, the CF was shown the avenue plantation raised along the Cheranmahadevi–Pathamadai road. Mostly tamarind seedlings had been planted, each surrounded by a mud wall of about four feet high to protect them from cattle. The seedlings, planted about two months earlier, had already put forth new shoots reaching 20-30 cm in height, thanks to regular watering using bullock carts. Seeing the healthy and well-maintained plants, the CF appeared satisfied.

He then proceeded to his headquarters at Tirunelveli, accompanied by the WLW. In spite of his stern and uncompromising nature with subordinates, the CF became calm and instructive when he observed sincere fieldwork carried out with real commitment. In those days, it was often said that if an inspection by the CF ended without any trouble, it was considered a great success. One could never expect open appreciation from such officers, but the absence of criticism itself was regarded as the highest form of praise.



V. Sundararaju, IFS., (R)

Former Deputy Conservator
of Forests

MY TERRITORIAL DEBUT

On quiet Sunday afternoon 12th March 1989, as I was relaxing at my home in Dharmapuri, one office assistant knocked the door and handed over an official letter from the District Forest Office Dharmapuri. To my surprise I found the posting orders to Dharmapuri territorial Range ordered by the then Salem Circle Conservator Thiru.A.S. Balanathan IFS.

Getting a postings to territorial Range was not easy those days, as it was the domain of veteran Rangers and I was also not very keen in selecting the Ranges. As soon as I finished my Range training in Attur Range, Salem district in 1982, I was posted to Chithar Rubber Division till 1985. My tenure in Rubber was enjoyable as I was Buildings Ranger and the Assistant Executive Engineer (AEE) who was also on deputation, taught me how to prepare buildings and roads estimate and to write measurement books. Life was enjoyable until one hypocrite Divisional Manager (DFO) took charge and started harassing me for about a year and I got transfer orders to Crash Plantation Division, Trichy on completion of three years. My happiness was short lived as the same DFO was also transferred and posted in Crash Plantation Division within three months.

To avoid further confrontation, I opted for deputation to Institute of Remote Sensing (IRS), Anna University Chennai. Though the officer regretted his actions in Rubber and tried to convince me I just ignored and joined as Scientist in IRS in the field of forestry and started learning about Remote Sensing techniques for about two years(1985-87). That was first time I saw the main frame computer in my life, processing the images from the data stored in magnetic tapes. We were engaged mainly in

cartography and not trained in Remote Sensing techniques. The then Director of IRS showed little interest in training the officers under deputation and that made me to revert back to department after a stint of two years though the deputation was for only one year.

I was posted as Survey and Demarcation Ranger and worked for about one year and transferred to Dharmapuri division, Wildlife Range, Pennagaram where there was a mini zoo in Hogenakkal. The wildlife range consists of only one forester to issue tickets to Crocodile Farm and some plot watchers to feed the animals. I was content in feeding the crocs and Otters until one fine afternoon I received the orders to take charge of the Dharmapuri territorial Range which I mentioned at the beginning of this article.



Range Office - Former Sir Thomas Munro Residence

But I didn't expect another news to follow next day while taking over charge. 13th March (my birthday) 1989, the head quarters Forester came to my house and escorted me to the big campus in the center of Dharmapuri town and I started taking stock of the new assignment. From a single Forester and few crocodiles, this was a whole new establishment with five foresters, ten Forest Guards,

Ten Forest Watchers, an office with Junior Assistant, Office Assistant, residential quarters with huge campus.

This Range office campus itself got some significance. Sir Thomas Munro, the then British Governor of Madras Presidency (1820-27) had stayed in this area from 1795 to 1798, while he was working as Assistant to the Superintendent of Revenue.

John Bradshaw in his book “Sir Thomas Munro, and the British settlement of the Madras Presidency” observed the following :



“writing of a spot near Dharmapuri, where he made a garden, Munro says that whenever he happened to be at Dharmapuri, he always spent at least an hour every day at it, 'and to quit it now goes as much to my heart as foresaking an old friend.'”

A pillar was erected by the British India's Government of Madras to mark the association of Dharmapuri with Sir Thomas Munro and the monument is few meters away from the Range office campus. It's also mentioned in the granite plaque, that the house and garden were few yards from the pillar site. Based on that note I presume that Sir Thomas Munro must have stayed in that Range office campus. Unfortunately few years back the Range office campus was taken over by Revenue officials for expanding the adjacent school/hospital and there is no trace of the old building now.

Coming back to 13th March 1989, on which date

I took charge of the Dharmapuri Range , it was not an easy task. Verifying the sandalwood seized under various offenses itself would take more than a week. While I was looking at the TCR, a telegram was received from the Conservator of Forests, Salem with a cryptic message saying “Sandalwood smuggling in Neruppur area. Seize and report”.

That was the first assignment on the very first day, even before I could take charge of the Range.. So myself and the Forester proceeded from Dharmapuri to Neruppur via Pennagaram in a two wheeler. It was around 3PM when we reached Neruppur and the Forest Guard and Watcher joined us and we proceeded by walk towards Yemanur. We have to go through the water spread area of the Stanley reservoir and it's a pretty open area with hills far away. As we were approaching the hill area we could see some movements of persons with back pack who are further moving into the jungle on seeing us. My subordinates then cautioned that they may belong to Veerappan's gang. Since we were only a few without any weapon , I decided to go back and seek some help.

I met the DFO in the night and explained the situation and requested him to speak to Superintendent of Police to get some additional force. Next day we again went to Neruppur with the local Sub-Inspector and three murmuring Constables with weapons. As we were moving towards the hills, we heard a gunshot which halted us on the track. The next shot squeezed past my right ear and we all ducked and lied flat on the ground. We were sure the shots were fired from the hill but couldn't see anybody. As we were lying down for about fifteen minutes there were no shots fired further. Myself and the Sub-Inspector were lying side by side and since there was no more gun shots we slowly tried to get up turned to see whether anybody hurt. To our surprise the Constables were not there and they have gone long before. Both

(Forest and Police) were not competent to combat in that kind of situation. So again I came back and met the DSP to arrange for more force and by the time we went to the same spot after a week nobody was there. Veerappan gang shifted to the unknown place deep in the jungle.

Later on Special Task Force was formed by the Police Department and I used to go with them whenever there is a combing operation in Neruppur, Yemanur and Hogenakkal areas. Karnataka cadre late Mr. P. Srinivas IFS, also used to visit my office in Dharmapuri and I have joined him in some combing operations from Hogenakkal to Anchetty in 1990. He was very simple, honest and down to earth person. I was in Chennai on court duty on that fateful day he was murdered by Veerappan.

Though there was not much natural regeneration of sandalwood in Dharmapuri Range, there are numerous network of roads connecting Bangalore and Andhra border for smuggling across the States. So our duty was mostly patrolling the roads all the time. There were no jeeps to patrol all the roads. So we used to hire a private car for night patrol and meet out all the expenditure unofficially. We need to bear all the expenditures like hired car, informants, equipment like spike rods torches etc. There was no provision for all these expenditures and somehow managed by all Rangers unofficially. The irony is now all the Rangers are provided with vehicles when there is no sandalwood stock that needs protection.

But in those adverse times, myself and my staff detected so many sandalwood offences and participated actively in many combing operations in Pennagaram area. After the execution of

Veerappan, Police department rewarded all the staff involved in the combing operations from the beginning. Whereas an Officer in our department made an arbitrary date and considered the staff for promotion from that date. I have led so many combing operations prior to that arbitrary date and he refused to entertain my application which had all the supporting documents like my weekly diary. The first three days of encounter with Veerappan gang, detailed several pages in my diary.

Exactly after three years of my stint in Dharmapuri Range, I was transferred to MSCS division in April 1992. I handed over the charge immediately and joined in Pennagaram MSCS Range and that should be a blessing in disguise. Had I continued in that Territorial Range another two months, I would have been asked to join the Vachathi raid and ceremoniously become an accused officer. Great escape!!!

So that was the only Territorial Range experience in my whole career.



**Author with
Rescued Panther
Cub from Palacode
Range, Dharmapuri
Division.
It was named "Pikkili"
after the Pikkili R.F,
where it was stranded**



K. Dhanapal, M.Sc.,

Former Deputy Conservator
of Forests

HELIOTROPIMUM INDICUM L. (BORAGINACEAE)

(தேள் கொடுக்கு பூண்டு)



Heliotropium indicum is commonly known in Tamil Nadu as Scorpion sting weed (தேள் கொடுக்கு பூண்டு). Other commonly used names are, Indian Heliotrope, Indian turnsole, Blue Heliotrope, Common Heliotrope, Wild Heliotrope, Devils weed etc. A flowering plant belonging to the family Boraginaceae, this species thrives both in tropical and subtropical regions, including parts of Africa, Asia, and the Americas. It is often found to colonize disturbed areas, roadside edges, and open fields, making it a pioneer species in various ecosystems.

Habitat and Distribution

An interesting aspect of this plant is that *Heliotropium indicum* behaves as either annual or perennial, depending on the growing conditions and climate. In its natural habitat in tropical and subtropical regions, it often thrives as a short-lived flowering perennial capable of regenerating year after year. While, in less favourable environments or harsher conditions, it tends to grow as an annual herb, completing its life cycle within a single year. *Heliotropium indicum* can grow up to 1 meter in height and has a sprawling habit. When conditions are optimal, it spreads quickly and tends towards being classified as weeds.

Taxonomy

Fragrant flowers of this plant, a typical corymb, are mostly white, light purple, or pale blue and feature a tubular shape with five lobes, attracting various pollinators such as bees and butterflies. The fruits of this plant are small, dry capsules containing many tiny hairy light weight seeds, capable of being dispersed by wind, animals, or water, contributing to the spread of this plant in its habitat. A single plant is capable of producing up to 2,500 seeds which are generally viable, in one growing season. The seeds that fall on the ground surface germinate easily. Indian Heliotrope seeds can also germinate and thrive in partially wet or flooded soils.

Propagation

Heliotropium indicum thrives in a warm climate for its optimal growth. This herbaceous plant can be propagated by seeds or by cuttings. It grows best in well-drained soil with moderate moisture. But, being a hardy plant, it can grow in

poor soils too. Regular monitoring is essential to control its spread, especially in areas where it may become invasive.

Uses

Heliotropium indicum is a versatile plant with many uses. This miracle plant is used as an ornamental plant in landscaping. Since its flowers attract bees and especially a large number of butterflies, it plays a vital ecological role in promoting pollination and increasing productivity of agriculture and horticulture crops. *H. indicum* is widely used in butterfly parks to attract, retain and sustain a good butterfly population. It has potential for use as a soil stabilizer and aid in controlling soil erosion.

Heliotropium indicum has been widely used in traditional medicine for its purported therapeutic properties. It is used in traditional system as an anti-inflammatory, analgesic and antimicrobial medicine. Various parts of this plant, including the leaves and flowers, are used to prepare poultices and infusions for treating wounds, infections, and other ailments.

As reported in Asian Journal of Pharmacognosy (2018) 2(1): 38-41, *Heliotropium indicum*, the whole plant or its parts are widely used by traditional medicine practitioners. Traditional healers in Kancheepuram, Tamil Nadu topically apply paste of whole plant to treat wounds and skin infections (Muthu et al., 2016). Healers in Kumaragiri Hills of Salem district, Tamil Nadu, apply leaf juice mixed with hot water to treat snake bite and scorpion sting (Alagesaboopathi, 2009).

In Mali, Africa, the plant is used for treatment of vomiting, amenorrhoea, ocular infections, and high blood pressure. In Nigeria and

Ghana, Sap of this plant is applied to gumboils, to clean ulcers, and to cure eye infections. This plant's dried Leaf powder is used for treating dermatitis, suppurating eczema and impetigo in children in Senegal. While in Indonesia, the leaf decoction is used in thrush and in poultices for herpes and rheumatism (reviewed in Togola et al., 2005). The list goes on. I have cited only a few.

Conclusion and a note of Caution

Heliotropium indicum is a versatile plant with both beneficial uses and potential risks. While *Heliotropium indicum* has medicinal uses, it can also pose risks. The plant contains *Tumorigenic Pyrrolizidine* alkaloids, that can harm livestock and humans when ingested in quantities and will cause symptoms like weakness, nausea, liver damage, or even death. Caution should be exercised when using this plant for medicinal purposes.

It is also considered an invasive weed that grows rapidly and can outcompete vegetation in diverse environments, affecting native biodiversity. Hence careful monitoring is needed.



V. Prabhakaran, IFS., (R)

Former Additional Principal
Chief Conservator of Forests

Dr. S. Paulraj, IFS., (R)

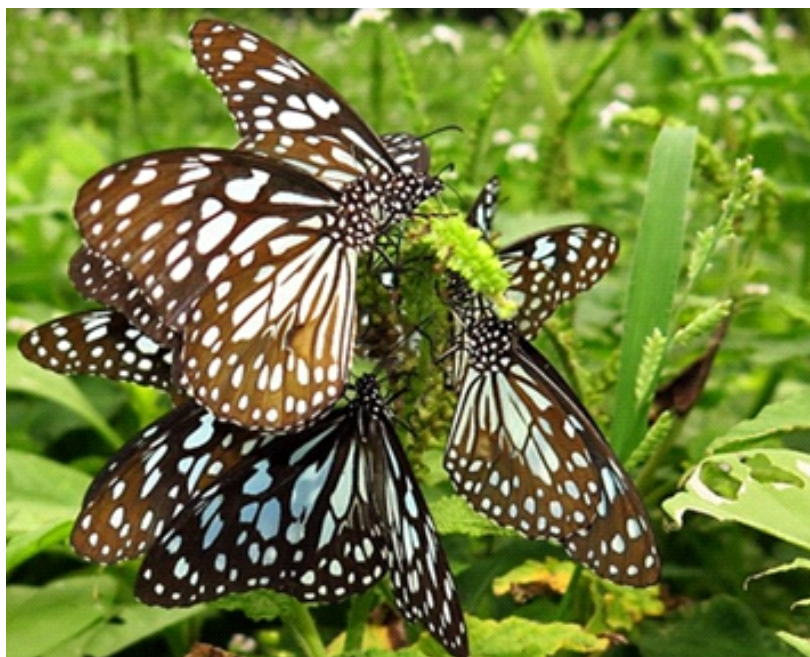
More interesting information on this fascinating plant species:

The so-called Tumorigenic Pyrrolizidine alkaloids present in this plant although can harm livestock and humans, is actually beneficial to the nectar butterflies as well as to this plant.

Studies indicate that danaid butterflies display a specialized association with *H. indicum* and use as nectar as well as alkaloid source. These butterflies use alkaloids for the production of pheromone precursors, facilitation of courtship behaviour and protection from their predators. The plant lures these butterflies to its nectar and alkaloids in order to be pollinated by them. Therefore, a mutual relationship exists between *H. indicum* and danaid adult butterflies. Another astonishing fact about this plant and butterflies' association is that, the butterflies feeding the plants parts are intoxicated by the alkaloids and rests and feed on the plants for a longer time there by aide effective pollination.

For further interesting information of this species, one may go through the following Research article by downloading from google search. It is worth reading.

Solomon Raju AJ, Suvarna Raju P. The coastal medicinally important herb species, Heliotropium indicum L. (Boraginaceae): pollination and danaid butterfly association.



Cluster of Butterflies feeding on *Heliotropium indicum*

சங்க இலக்கியத்தில் இயற்கை உவமை (குறுந்தொகை)

நாம் ஒரு பொருளை அல்லது ஒரு உயிரை பார்க்கும்போது அது நமக்குத் தெரிந்தால் கடந்து சென்று விடுகிறோம். அதன் பெயர் தெரியவில்லை என்றால் முயற்சி செய்து தெரிந்து கொள்வோம் இல்லையேல் விட்டு விடுவோம். உதாரணமாக ஒரு செடியில் அழகான மலர் ஒன்று இருந்தால் ஓர் இலக்கியவாதி அதை கண்டு ரசித்து கவிதை அல்லது கட்டுரை எழுதுவான். ஒரு தாவர ஆய்வாளர் அதை பார்த்தால் அதை பிரித்து ஒவ்வொரு பாகத்தையும் என்னவென்று ஆராய்ச்சி செய்வான்..ஒரு பக்தன் அதை இறைவனுக்கு தூடி மகிழ்ந்து வணங்குவான்.

நோபல் பரிசு பெற்ற அறிஞர் பெர்னாட்ஷா தன் வீட்டை சுற்றி அழகான செடி கொடிகளையும் மலர்களையும் வளர்த்து வந்தார் ..அவர் ஒரு பூவை கூட பறித்தது கிடையாது. அதனால் அவர் வீட்டில் மலர் குடுவை கிடையாது. ஒரு நண்பர் அவர் வீட்டிற்கு வந்திருந்தார் அழகான பூக்கள் இருக்கின்றனவே மலர் குடுவையில் வைக்கலாமே என்று அறிவுரை கூறினார் .அதற்கு பெர்னாட்ஷா எனக்கு சிறிய குழந்தைகள் மேல் அளவு கடந்த ஆசை அதனால் அந்த குழந்தைகளின் கழுத்தை வெட்டிக்கொண்டு வீட்டில் வைக்கலாமா? என்று பதில் கூறினாராம்.. மலர்களை அவர் குழந்தைகளுக்கு சமமாக எண்ணினார் .

விலங்கியலின் தந்தை என்று கருதப்படும் கிரேக்க நாட்டு மேதை அரிஸ்டாட்டில் அவர் உயிரினங்களை ஆராய்ச்சி செய்து விலங்கினங்களை அறிந்து கொள்ளும் முறையை உருவாக்கினார் . அவர் ஒரு பொருளை அல்லது உயிரை எப்படி ஆவணப்படுத்த வேண்டும் என்று விதிமுறைகள் வகுத்துள்ளார். அதை (Category of understanding) என்பர். அதை நாம் தெரிந்துகொள்வது அவசியம் .. விவரம் பின்வருமாறு :

1. SUBSTANCE - பொருள் 2. QUANTITY - அளவு 3. QUALITY - குணம் 4. PLACE - இடம் 5. TIME - காலம் 6. RELATION - உறவு 7. POSITION - நிலை 8. STATE - நிலைப்பாடு 9. ACTIVITY - செயல் 10. PASSIVITY - விளைவு ஆகவே ஒன்றை ஆவணப்படுத்தும் போது மேல் கூறியது பின் பற்றவேண்டும்.

மேற்கூறிய முறைகளைப் நமது முன்னோர்கள் ஆயிரம் ஆண்டுகளுக்கு முன்பு உயிர் உள்ளவற்றையும் , உயிர் அற்றவைவற்றையும் வகைப்படுத்தி , பெயரிட்டு அதன் குணம் ஆகியவற்றை கண்டறிந்து . பாடியுள்ளனர் . . . சுருக்கமாக சொல்வதென்றால் இயற்கையோடு ஒன்றி வாழ்ந்தனர்.. தாவரங்களைக் கூட உடன் பிறப்பு என்று கருதினர். . இதை நாம் அடிக்கடி கேட்கிறோம் “ வாடியப் பயிரைக்கண்டு வாடினார் வள்ளலார் “ . . நாம் ஒன்றை இந்த இடத்தில் தெரிந்து கொள்ள வேண்டும் .அறிதல் என்பது வேறு உணர்தல் என்பது வேறு . நமது புலவர்கள் எவ்வளவு கூரிய நுண்ணறிவு கொண்டவர்கள் என்பது அவர்கள் காட்டிய உவமைகளை வைத்து அறிய முடியும் .

சங்க இலக்கியத்தில் உவமைகள் அதிகம் காணப்படுவது எட்டுத்தொகை நூல்களான குறுந்தொகையாகும். உவமை என்பது ஒன்றை விளக்க தெரிந்த வேறு ஒன்றை கூறி புரிய வைப்பது.. இது அடிதட்டு மக்களுக்கும் புரியும் கலை நயம் இருக்கும்.. குறுந்தொகை 401 பாடல்களைக் கொண்டது இதை 205 புலவர்கள் இயற்றி உள்ளனர் . தொல்காப்பியத்திற்கு உரை எழுதிய ஆசிரியர்கள் குறுந்தொகையின் 235 பாடல்களை மேற்கோளாக எடுத்துக்காட்டி உள்ளனர் எனபது குறிப்பிடத்தக்கது

இனி நமக்கு பயன் படும் வகையில் தாவரங்களையும் விலங்குகளையும் ஒப்பிட்ட சில உவமைகளை காண்போம்... பாடல்களை விரிவாக பார்க்காமல் அதன் சாரத்தை மட்டும் பார்க்கலாம்.

1. பொற்கொல்லர் பொன்னுரைத்த கல் குயிலின் இறகில் பூவின் மகரந்த தாது படிந்திருப்பது போன்று இருக்கும்.

“மின்னின் தூவி இருங்குயில் ,பொன்னின் உரை திகழ் கட்டளை கடுப்ப”

(பாடல் : :192 பாடியவர் : கச்சிப்போட்டு நன்னாநாகையார்)

2. தாழம்பூமுள் (*Pandanus odoratissimus*,

Common Name: Screw pine flower)

அணிலின் பல் போல் இருக்கும்

சம்மந்தப்பட்ட பாடல் வரிகள்

“ அணில் பல் அன்ன கொங்கு முதிர் முண்டகத்து “

(குறுந்தொகை- பாடல் :49 நெய்தல்.

பாடியவர் -அம்முவானார்,



3. ஞாழல் மலர் (புலி கொன்றை)

(*Cassia sophora* - Common name -Tiger claw tree)

சிறு கடுகு போல் இருக்கும்.

சம்மந்தப்பட்ட பாடல் வரிகள்

“ஐயவி அன்ன சிறுவீ ஞாழல் “

(பாடல் :50 :. பாடியவர்- குன்றியனார்)



4. புங்கை மலர் (*Pongamia pinnata*.

common name: Pongamia tree)

அரிசி பொறி போன்றிருக்கும்
சம்மந்தப்பட்ட பாடல் வரிகள்
“நனை முதிர் புன்கின் பூத்தாழ் வெண்மணல்
“செந்நெல் வான் பொரி சிதறி அன்ன”
(பாடல் :53, பாடியவர்: கோப்பெருஞ்சோழன்)



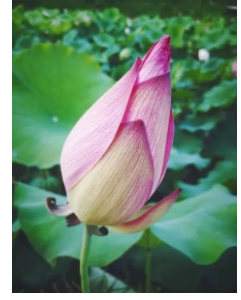
5. கூம்பிய ஆம்பல் மலர்கள் (*Nymphae nouchali* ,

Common name: Water lily) கொக்கின் முதுகு போல்

தோன்றும் சம்மந்தப்பட்ட பாடல் வரிகள்

“பைங்காற் கொக்கின் புன்புறத்தன்ன
குண்டு நீர் ஆம்பலும் கூம்பின;”

(பாடல் :122 - பாடியவர் ஓரம் போகியார்)



6. கற்களின் மேல் வேங்கை மலர் (*Pterocarpus marsupiam*.

Common name : Indian kino tree)

உதிர்ந்து கிடக்கும். அவை புலிக்குட்டிகள்

படுத்திருப்பது போன்று தோன்றும்

சம்மந்தப்பட்ட பாடல் வரிகள்

“கருங்கால் வேங்கை வீசு துறுகல்

இரும்புலிக் குருளையின் தோன்றும் காட்டிடை”

(பாடல்: 47 , பாடியவர்: நெடுவேண்ணிலவினார்)



7. கற்களின் மேல் காந்தள் மலர் (*Gloriosa superba*,

Common name : Flame Lily, Glory lily) பூத்துக் கிடக்கும்

அவை செம்புள்ளிகள் நிறைந்த யானையின் முகம்
போல் தோன்றும்.

“பொருத யானைப் புகர் முகம் கடுப்ப

மன்றத் துறுகல் மீமிசைப் பல உடன் “

(பாடல் ;284, பாடியவர்: மிளைவேள் தித்தன்)



8. வாகை மலர் கொத்து (*Albizia lebbbeck*. common name:

Sari's tree) மயிலின் கொண்டையை போன்றிருக்கும்

சம்மந்தப்பட்ட பாடல் வரிகள்

“குமரி வாகைக் கோல் உடை நருவி “

(பாடல் 347 - பாடியவர் சேந்தன் கண்ணனார்)



9 பூத்த காய மலர் (*Memecylon umbellatum*, common name: Common name:

Iron wood tree flower) மயில் கொண்டையைப் போல் இருக்கும்

சம்மந்தப்பட்ட பாடல் வரிகள்

“புல்லென காயாப் பூக்கொழு பெருஞ்சினை
மென்மயில் எருத்தின் தோன்றும் “

(பாடல்: 187, பாடியவர்: ஒளவையார்)



10. சேம்பின் இலை (Botanical name: *Colocasia esculenta*. common name: Tara leaves) யானையின் காதை போன்றிருக்கும்

சம்மந்தப்பட்ட பாடல் வரிகள்

“சிலம்பிற் சேம்பின் அலங்கல் வள் இலை
பெருங்களிற்றுச் செவியின் மானத் தைஇ “

(பாடல்: 76 - பாடியவர்: கிள்ளி மங்கலன் கிழார்)



11. நொச்சி இலை (*Vitex negundo*. Common name: Chase tree)

மயிலின் காலடியைப் போன்றிருக்கும்

சம்மந்தப்பட்ட பாடல் வரிகள்

“மயில் அடி இலைய மாக்குரல் நொச்சி “

(பாடல் :138, குறிஞ்சி, பாடியவர்: கொல்லன் அழிசி)



12. அடும்பின் (*Ipomoea pes-caprae* (L.) R.Br., common name: Beach morning glory) இலை மானின் அடியை போன்று இருக்கும்.

சம்மந்தப்பட்ட பாடல் வரிகள்

“மான் அடி அன்ன கவட்டிலை அடும்பின் “

(பாடல்: 243, ந, பாடியவர்: நம்பி கூட்டுவான்)



13. பாறையிலிருந்து வீழும் அருவி பாம்பின் சட்டையை போன்றிருக்கும்
சம்மந்தப்பட்ட பாடல் வரிகள் “ஓம்பு மதி வாழியோ வாடை - பாம்பின்
தூங்கு தோல் கடுக்கும் தூவெள் அருவி” (பாடல்: 235 பாலை பாடியவர்:
மாயந்தன் அறிவினர்)

14. அருவி நீர் பாறை. கற்கள் இடையே செல்லும்போது பாம்பு நிலத்தில்
ஊர்வது போல் இருக்கும். சம்மந்தப்பட்ட பாடல் வரிகள் “கல் பொருது இரங்கும்
கதழ் வீழ் அருவி நிலம் கொள் பாம்பின் இழி தரும்” (பாடல் 134 பாடியவர் :
கோவேங்கை பெருங்கதவன்)



சங்க காலத்தில் தமிழ் புலவர்கள் இயற்கையை கண்டு ரசித்தது அல்லாமல் அதை ஒட்டி உறவாடி வாழ்ந்திருக்கின்றனர் . அவர்களுக்கு பல முகங்கள் , புலவர், கவிஞர், ஆராய்ச்சியாளன் இன்னும் பல., ஒவ்வொரு உயிரையும் ஒன்றோடு ஒன்று ஒப்பிட்டு அதை சாதாரண மக்களுக்கும் புரியுமாறு கூறியுள்ளது நமது தமிழ் மொழிக்கு சிறப்பாகும்.



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முன்னாள் மேலாளர்
பாரத ஸ்டேட் வங்கி

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5. Thiru Venkatasamy, DCF., (R)
6. Thiru P. Raja Muthiayan, IFS., APCCF (R)

கடல் வளம் காக்கும் பவளப்பாறைகள்!

பவளப்பாறைகள் 'கடலின் மழைக்காடுகள்' என்று அழைக்கப்படுகின்றன. பவளப்பாறைகள் உயிர்க்கோளத்திற்கும், மனிதர்களுக்கும் மிகவும் முக்கியமானவை. இவை பார்ப்பதற்கு மிக அழகாகக் காட்சி அளிக்கும். பவளப்பாறைகள் கடற்கரைப்பகுதிகளுக்குப் பாதுகாப்பைக் கொடுக்கின்றன. உணவிற்காகவும், மருந்திற்காகவும் பயன்படும் மீன்கள், சிப்பிகள் மற்றும் சங்குகள் போன்றவற்றிற்கு இவை புகலிடமாக அமைகின்றன.

பவளங்கள் ஏறத்தாழ 542 மில்லியன் ஆண்டுகளுக்கு முன்னாள் காம்ப்ரியன் 'Cambrian' காலத்தில் உருவானதாக அறிவியல் ஆய்வுகள் தெரிவிக்கின்றன. உலக அளவில் கடல்பகுதிகளில் உள்ள பவளப்பாறைகளில் சுமார் 10% உயிரிழந்த நிலையில் காணப்படுகின்றன. சுமார் 60% பவளப்பாறைகள் மனிதர்களின் பல்வேறு செயல்பாடுகளால் ஆபத்துக்குள்ளாகி இருக்கின்றன. குறிப்பாக தென் கிழக்கு ஆசியக் கடல்பகுதிகளில் சுமார் 50% பவளப்பாறைகள் மிகவும் ஆபத்தான நிலைக்குத் தள்ளப்பட்டுள்ளன. 2030-ஆம் ஆண்டு வாக்கில் உலக அளவில் காணப்படும் பவளப்பாறைகளில் சுமார் 50% அழிந்துவிடும் என்று அஞ்சப்படுகின்றது.

பல்வேறு மாசுக்கள் (வீட்டுக் கழிவு, பிளாஸ்டிக் மாசு, தொழிற்சாலை மாசு, உப்பளங்கள், எண்ணெய் மாசு முதலானவை), அளவுக்கு அதிகமாக மீன்பிடித்தல், அழிவை ஏற்படுத்தும் மீன்பிடி முறைகள் (வெடி வைத்து மீன் பிடித்தல், விஷத்தைப் பாய்ச்சி மீன்பிடித்தல், கன்னி வைத்து மீன் பிடித்தல், ஒலைவலை, சூழ்வலை மற்றும் இழுவலையைப் பயன்படுத்துதல்), உயிருள்ள பவளங்களைச் சேகரித்தல், பவளப்பாறைகளை வெட்டிஎடுத்தல் ஆகிய பல்வேறு காரணங்களால் பவளப்பாறைகள் மிகவும் ஆபத்துக்குள்ளாகி வருகின்றன. அதனால், உலகின் பல்வேறு நாடுகளும் இந்த அரியவகைப் பவளப்பாறைகளைப் பாதுகாப்பதற்கான தீவிர நடவடிக்கைகளில் ஈடுபட்டுள்ளன.

பூமியில் நிலவும் பல்வேறு விதமான சூழியல் தொகுதிகளில் பல வகையான உயிரினங்களைத் தன்வசம் கொண்டுள்ள பவளப்பாறைகள் மிக முக்கியமானதாகும். ஆயிரக்கணக்கான உயிரினங்கள் அவைதம் உணவு, உறைவிடம் மற்றும் பாதுகாப்பிற்குப் பவளப்பாறைகளையே சார்ந்து வாழ்கின்றன.



கடல்வாழ் உயிரினங்களின் வாழ்விடமாகத் திகழும் பவளப்பாறைகள் பவளப்பாறைகள் அதிக அளவு உயிரின உற்பத்தி மையங்களாக விளங்குகின்றன. கரியமிலவாயுவைச் சேமித்துவைத்து உயிரினங்கள் வாழ் மிகச் சிறந்த சேவை புரிகின்றன. கடற்கரை ஓரம் அரிமானம் ஆகாமல் தடுக்கின்றன. புற்று நோய், AIDS போன்ற நோய்களுக்குத் தேவைப்படும் மருந்துகள் தயாரிக்க மூலப்பொருட்கள் பவளப்பாறைகளிலிருந்து கிடைக்கின்றன. எலும்பு முறிவு சிகிச்சை மற்றும் எலும்பு சம்பந்தமான நோய்களுக்கு இவை மருந்தாக விளங்குகின்றன. இந்திய பாரம்பரிய மருத்துவத்திலும் பவளங்கள் முக்கிய பங்கு வகிக்கின்றன.

இந்தப் பவளப்பாறைகள் உருவாகின்ற விதம் மிகவும் அதிசயமானதாகும். பவளங்கள் குழிஉடலிகள் என்ற உயிரினத்தைச் சேர்ந்தவை. பெரும்பாலான பவளங்கள் கூட்டமாக ஒன்றாகச் சேர்ந்து வாழும். தனித்து வாழக்கூடிய பவளங்களும் இருக்கின்றன. நூற்றுக்கும் மேற்பட்டு, கணக்கற்ற பவளப்பாலிப்புகள் ஒன்றாக இணைந்து, கூட்டமாக வாழக்கூடிய பவளங்களை உற்பத்தி செய்கின்றன.

பவளப் பாலிப்புகளின் வயிற்றுப் பகுதி ஒருபுறம் மட்டுமே திறக்கும்படியான அமைப்பினைக் கொண்டது. வயிற்றின் திறந்திருக்கும் முனைப் பகுதியே இதன் வாய் எனப்படுகின்றது. இந்த வாய்ப் பகுதியைச் சுற்றிலும் tentacles எனப்படும் நூலிழை போன்று உணர்கொம்புகள் உள்ளன.

இந்தப் பாலிப்புகள் மாமிச உண்ணிகள். இவை தம் உணர்கொம்புகளைக் கொண்டு அவற்றுக்கான இரையைப் பிடிக்கவும், எதிரிகளிடமிருந்து தற்காத்துக் கொள்ளவும், கழிவுகளை வெளியேற்றவும் செய்கின்றன. இதன் இரை வாய் வழியாக வயிற்றுப் பகுதியை அடைகின்றது. உணவு செரித்தவுடன் அவை மீண்டும் இரையைப் பிடித்து உண்ணத் தொடங்கிவிடும். மிதவை உயிரினங்களை உணவாகக் கொள்ளும் பவளப்பாலிப்புகள் பெரும்பாலும் இரவு நேரங்களிலேயே இரை தேடும் பணியில் ஈடுபடுகின்றன. பவளங்கள் கடல் நீரிலே நீந்தி வாழும் புழுவாக இருக்கும்போதே கண்டங்கள் மற்றும் தீவுகளின் ஓரங்களில் நீரில் மூழ்கியிருக்கும் பாறைகள் அல்லது வேறு கடினமான பரப்புகள் மேல் அவற்றைப் பற்றி வளரத் துவங்கும்.

பவளங்கள் விரிவடைந்து வளரும்போது பவளப் பாறைத் திட்டூக்கள் உருவாகின்றன. இவை வளம் செறிந்த உயிரியப் பன்மையத்தையும், பலவித இயற்கை வாழ்விடங்களையும் தன்னகத்தே கொண்டு விளங்குவதோடு மிகவும் தொன்மை வாய்ந்ததாகவும் கருதப்படுகின்றது. பவளங்கள் பவளப்பாறைத் திட்டூக்களாக உருவாக ஏறத்தாழ 1,00,000 முதல் 30 மில்லியன் ஆண்டுகள் வரை ஆகும் என்று அறிவியல் ஆய்வுகள் கூறுகின்றன.

காலி ப்ளவர் பவளப்பாறை (Cauliflower coral), மான்கொம்பு பவளப்பாறை (Staghorn coral), பெருவிரல் பவளப்பாறை (Finger coral), மேடைவடிவ பவளப்பாறை (Table coral), கள்ளிவடிவ பவளப்பாறை (Cactus coral), தலைவடிவ பவளப்பாறை (Head coral), மூளை வடிவ பவளப்பாறை (Brain coral), தட்டு வடிவ பவளப்பாறை (Plate coral), படர் பவளப்பாறை (Merulina coral), கொத்து பவளப்பாறை (Rasp coral) தூண்வடிவ பவளப்பாறை (Pillar coral) என்று பவளப்பாறைகள் அவற்றின் உருவ அமைப்பைக் கொண்டு பல்வேறு பெயர்களில் அழைக்கப்படுகின்றன.

பவளப்பாறைகள் பூமத்திய ரேகைக்கு அருகில் உள்ள மித வெப்பமண்டல நாடுகளின் கடற்கரைப் பகுதிகளில் அதிகமாகக் காணப்படுகின்றன. மத்திய இந்திய பசிபிக் மற்றும் கரீபியன் பகுதிகள் அதிக உயிர் இனங்களை உள்ளடக்கிய பளப்பாறைகளைக் கொண்டவையாகும்.

உலகில் உள்ள பவளப்பாறைகளில் சுமார் 60% இந்தியப் பெருங்கடலில் காணப்படுகின்றது. இந்தியாவில் உள்ள பவளப்பாறைகளின் பரப்பளவு சுமார் 2375

சதுர கி.மீ ஆகும். இந்தியாவைப் பொறுத்தவரையில் அந்தமான் நிகோபார் தீவுகள், லட்சத் தீவுகள், மன்னார் வளைகுடா, பாக் ஜலசந்தி, கச்ச வளைகுடா ஆகிய பகுதிகளில் பவளப்பாறைகள் அதிகம் காணப்படுகின்றன.

ஆஸ்திரேலியாவின் குயின்ஸ்லாந்து மாநிலக் கடற்கரையில் அமைந்துள்ள 'கிரேட் பாரியர் ரீப்' என்று அழைக்கப்படும் 'பேரரன் பளப்பாறைத் திட்டு' உலகிலேயே மிகப்பெரிய பளப்பாறைத் திட்டாகக் கருதப்படுகிறது. சுமார் 2500 கிலோ மீட்டர் நீளமும், சுமார் 15-முதல் 350-கிலோ மீட்டர் அகலமும் கொண்டது. சுமார் 2500-க்கும் அதிகமான சிறிய திட்டுக்கள், காயல்கள் மற்றும் தீவுகளைத் தன்னகத்தே கொண்ட இந்தப் பளப்பாறைத் திட்டு ஏறத்தாழ 2,25,000-சதுர கிலோ மீட்டர் பரப்பளவில் பரந்து விரிந்துள்ளது.

பளப்பாறைகளின் முக்கியத்துவத்தைக் கருத்தில் கொண்டு வன உயிரினப் பாதுகாப்புச் சட்டம்-1972 இல் அட்டவணை 1 இன் கீழ் கொண்டுவரப்பட்டு அவைகளுக்கு உரிய பாதுகாப்பு வழங்கப்பட்டுள்ளது. மன்னார் வளைகுடாவில் உள்ள வான் தீவில் செயற்கை முறையில் உருவாக்கப்பட்ட கான்க்ரீட் பவளப்பாறைகளை அங்கே அறிமுகம் செய்து வளர்த்த பிறகு, அதன் பரப்பளவு சுமார் 2-ஹெக்டர் அதிகரித்துள்ளது கண்டறியப்பட்டுள்ளது. அதே போன்று பிற தீவுகளிலும் செயற்கைப் பவளப்பாறைகள் அறிமுகப் படுத்தப்பட்டு வருகின்றன.

இப்படிக் கடல்வாழ் உயிரினங்களின் வாழ்விடமாகத் திகழ்ந்து இயற்கைச் சமநிலையைப் பேணி வரும் சூழியல் தொகுதியாகிய பவளப்பாறைகளைப் பாதுகாப்பது நம் கடமை அல்லவா!



வ.சுந்தராஜ் இ.வ.ப (இ)

முன்னாள் துணை
வனப்பாதுகாவலர்

வையம் வளம் பெற வனங்களைக் காப்போம்

ஒரு நாட்டின் பொருளாதாரம் நிலைத்திருக்க வேண்டுமென்றால் அந்த நாட்டின் இயற்கை வளம் நிறைந்த வனப் பகுதிகள் நல்லபடியாகப் பாதுகாக்கப்பட வேண்டும். ஏனெனில் ஒரு நாட்டில் வாழும் மக்களின் சமூகம் மற்றும் பொருளாதாரம் மேம்பாடு அடைய வேண்டுமெனில் அதற்கு ஆதாரமாக விளங்கும் உயிரியப் பன்மையின் உறைவிடமாகத் திகழும் வனங்கள் முறையாகப் பேணப்பட வேண்டும். இவை மட்டுமல்லாது ஒரு நாட்டின் இயற்கை அரணாகவும், தட்ப வெப்ப நிலைகளைச் சீர் செய்யும் சக்தியாகவும் வனங்களே திகழ்கின்றன. அதனால்தான் ஈராயிரம் ஆண்டுகட்கு முன்னரே வள்ளுவப் பெருந்தகை "மணி நீரும் மண்ணும் மலையும் அணி நிழற் காடும் உடையது அரண்" என்று வனத்தின் பெருமையினைப் பறை சாற்றியுள்ளார்.

இந்த மண்ணுலகில் வாழும் அனைத்து உயிர்களுக்கும் ஆதாரமாக விளங்கி அவர்தம் நலம் நிறைந்த வளமான வாழ்க்கைக்குத் தேவையான அனைத்துப் பொருட்களையும் வழங்கி வருகின்ற காரணத்தால் வனங்கள் என்றும் "அமுதசுரபி" எனப் போற்றப்படுகின்றன. வனங்கள் பல்வேறு வகையான உணவுப் பொருட்களையும், மருந்துப்பொருட்களையும் மற்றும்

மரங்களையும் வழங்குவதோடு தொழிற் சாலைகளுக்குத் தேவையான மூலப் பொருட்களையும் தந்து உதவி அதன் மூலம் மக்களுக்கு வேலை வாய்ப்பையும் உருவாக்கிக் கொடுத்துள்ளன. மலை வாழ் மக்களின் வாழ்விடமாகவும் உணவுக் களஞ்சியமாகவும் விளங்கி வருபவை வனங்களே.

மண்ணையும் விண்ணையும் இணைக்கும் அற்புதப் படைப்பாகத் திகழ்பவை வனங்களே. பலகோடி மைல்களுக்கு அப்பால் உள்ள ஆதவன் உமிழ்கின்ற சக்தி, அண்டவெளியிலுள்ள கரியமிலக் காற்று, மண்ணின் ஆழ் மட்டத்திலுள்ள பல்வேறு மூலகங்கள் மற்றும் மழை நீர் இவற்றையெல்லாம் ஒன்று சேர்த்து அனைத்து உயிரினங்களுக்கும் தேவையான உணவையும், மற்றும் உலகிற்குப் பயன் அளிக்கக் கூடிய பல்வேறு பொருட்களையும் உருவாக்கும் ஆற்றல் பெற்றுத் திகழும் மரங்கள் நிறை வனங்களே நம் வாழ்வின் அடிப்படை ஆதாரமாகத் திகழ்கின்றன. மண் அரிப்பு மற்றும் மண் சரிவைத் தடுத்து மண் வளத்தைப் பாதுகாப்பதோடு, நீர் சுழற்சியையும் பராமரித்து சூழியலின் தரத்தை எப்போதும் காத்து வருபவை வனங்களே.

அனைத்து உயிர்கட்கும் அடிப்படைத் தேவையான உயிர் வளியை வெளியிட்டு, கரியமில் வாயுவைக் கிரகித்து, மழை நீரைச் சேகரித்து நமக்குத் தேவையான நீரை வழங்கி, மண் வளத்தைப் பேணி, வேலை வாய்ப்பை வழங்கி எல்லாவற்றுக்கும் மேலாக பூமியின் வெப்ப உயர்வைக் கட்டுப்படுத்தும் மகத்தான சேவைகளை ஆற்றிவருபவைக் காடுகளே.

ஆனால், காலப் போக்கில் காடுகளை நம்பி வாழும் ஏழ்மை நிலை மக்களின் தேவைகளாலும், பண்பலம் படைத்தவர்களின் பேராசையினாலும், சமூக விரோத சக்திகளாலும் வனங்கள் அழிக்கப்பட்டு வருகின்றன.

ஆகவே, உலகம் உயிர்த்திட இந்தப் பூமியில் வாழும் உயிரினங்களின் வாழ்வாதாரமாகத் திகழும் வனங்களைப் பாதுகாக்க வேண்டும் என்ற கொள்கை அடிப்படையில் ஒவ்வொரு ஆண்டும் மார்ச், 21 ஆம் தேதியினை உலக வன நாளாக உலகில் உள்ள ஒவ்வொரு நாடும் சிறப்பாகக் கொண்டாடி வருகின்றன. இந்த நாளில் சூரியன் பூமத்திய ரேகையைக் கடந்து செல்வதோடு, வசந்த காலத்தை வரவேற்பதாகவும் இந்த நாள் அமைகின்றது. அத்தோடு, இது வறட்சியான காலமாக இருந்தபோதிலும், இந்தச் சமயத்தில்தான் தாவர இனங்களிடையே அதிகமான வளர்ச்சி காணப்படுவதும் இன்னாளுக்குரிய தனிச் சிறப்பாகும்.



வன்னி மரங்களைக் காப்பதற்காக 363-பீஷ்னாய் மக்கள் உயிர்த்தியாகம் செய்த வீரவரலாறு

வனங்களைக் காப்பாற்றும் பொருட்டு, உயிர்த் தியாகம் செய்தவர்களைப் பற்றியும், மரங்களின்பால் மாறாத பற்றுக் கொண்டவர்களைப் பற்றியும் விரிவாக எடுத்துரைக்க வேண்டும். கி.பி.1730 ஆம் ஆண்டு ராஜஸ்தான் மாநிலத்தில் 'கெஜார்லி' கிராமத்தில் தாம் வளர்த்த வன்னி மரங்களைக் காப்பதற்காக வேண்டி 363 பீஷ்னாய் இன மக்கள் உயிர்த் தியாகம் செய்த வீர வரலாற்றை மாணவர்களிடையே எடுத்துரைக்க வேண்டும். அமெரிக்காவின் பூர்வ குடிகள் செவ்விந்தியர்கள் என்பது நம் எல்லோருக்கும் தெரியும். அந்நாட்டின் பெரும்பகுதி நிலப்பரப்பு செவ்விந்தியர்கள் வசம் இருந்தது. அமெரிக்க அரசு பெரிய நிலப்பரப்பை அவர்களிடம் இருந்து விலைக்கு வாங்க விரும்பியது. அதற்காக வேண்டி 1854 ஆம் ஆண்டு அமெரிக்க அரசாங்க அதிகாரிகள் செவ்விந்தியர்களின் தலைவனாக விளங்கிய 'சியாட்டில்' (Seattle) அவர்களைச் சந்தித்தனர். அவரிடம்

அரசின் வேண்டுகோளைத் தெரிவித்தனர். நிலத்தைக் கேட்பதோ அரசாங்கம். அதை மறுத்தாலும் அவர்களால் அவர்தம் நிலங்களைப் பாதுகாக்க முடியாது என்பதை நன்கு அறிந்திருந்தார் சியாட்டில். இருந்தபோதும், அவர்தம் அன்னை பூமியின் மீது அவருக்கிருந்த பாசம் அவரைக் கீழ்க் கண்டவாறு பேசத் தூண்டியது.



சியாட்டிலின் உருவப்படம்

சியாட்டிலின் சிந்தை கவர்ந்த உரை: "பூமி மனிதருக்குச் சொந்தமானது அல்ல; மனிதர்கள் தான் பூமிக்குச் சொந்தமானவர்கள். இந்த பூமிக்கு என்ன நேர்கிறதோ, அது பூமியின் குழந்தைகளாகிய மனிதருக்கும் நேரும்.

உயிர்வலை மனிதரால் பின்னப்பட்டதல்ல; அதில் மனிதன் ஓர் அங்கம் மட்டுமே. பூமியே எங்கள் தாய். நம் காலடியின் கீழ் உள்ள நிலம் எங்கள் முன்னோர்களின் சாம்பல். பூமியை நமது மூதாதையர்களிடம் இருந்து நாம் சுவீகாரமாகப் பெறவில்லை; இதை நமது குழந்தைகளிடம் இருந்து கடனாக வாங்கியுள்ளோம். ஓடைகளிலும், ஆறுகளிலும் ஓடுவது வெறும் தண்ணீர் மட்டுமல்ல; அது எங்கள் முன்னோரின் இரத்தமாகும். ஆறுகள் எங்கள் சகோதரர்கள்; அவர்கள்தான் நமது தாகத்தைத் தணிக்கின்றனர். காற்று மிகவும் விலைமதிப்பு மிக்கது; பூமியில் உயிர்த்திருக்கும் விலங்கினங்கள், தாவரங்கள் மற்றும் மனிதர்கள் ஆகிய யாவரும் காற்றைப் பங்கிட்டுக் கொள்கின்றனர். விலங்குகள் இல்லாமல் மனிதரால் வாழமுடியுமா? பூமியில் வாழும் அனைத்து விலங்கினங்களும் மறைந்து போனால், அந்த மயான அமைதியிலேயே மனிதன் மரித்துப் போவான். இங்கே வாழும் அனைத்தும் ஒன்றோடு ஒன்று தொடர்புடையவை".

இவ்விதம் உரை ஆற்றியபோதும் தமது நிலங்களை அமெரிக்க அரசுக்கு வழங்க இறுதியில் ஒப்புக்கொண்டார். இப்படி ஓர் உருக்கமான உரையினைக் கேட்ட அமெரிக்க அதிகாரிகள் மனம் நெகிழ்ந்து போனார்கள். பூமியின் மீதும், பிற உயிர்கள் மீதும் அவருக்கிருந்த ஆழமான பாசத்தைக் கண்டு பிரமித்துப் போயினர். சியாட்டில் அவர்களைப் பற்றி

அமெரிக்க அரசிடம் எடுத்துரைத்து அவருக்கு உரிய மரியாதை செய்ய விரும்பினார்கள். அதன்படியே அங்கே உருவாக்கப்பட்ட நகருக்கு அவர் பெயரையே 'சியாட்டில்' என்று சூட்டியதோடல்லாமல், அவரின் நினைவைப் போற்றும் வண்ணம் அவரது முழு உருவச் சிலை எழுப்பி அவருக்கு மனமுவந்து மரியாதை செய்தனர். இப்படி இயற்கையின் மீது மாறாக் காதல் கொண்ட மாமனிதர்களைப் பற்றி இன்றைய தலைமுறையினர் அவசியம் தெரிந்துகொள்ளவேண்டும்.

1973 ஆம் ஆண்டில் உத்தர்காண்ட் மாநிலத்தில் 'சிப்கோ இயக்கம்' தோன்றியது. பின்னர் இந்த இயக்கம் 'சுந்தர்லால் பஹுக்ஷா' என்ற சுற்றுச் சூழல் ஆர்வலரின் தலைமையில் தீவிரமாகச் செயல்படத் தொடங்கியது.

ஜாதவ் பாயங் மொலாய்' என்ற அஸ்ஸாம் மாநில ஆதிவாசி, தனி ஒரு மனிதனாய் 1360 ஏக்கர் காடுகளை உருவாக்க முடிந்தது என்றால், ஏன் நாம் ஒன்றிணைந்து செயல்பட்டால் அது போன்ற காடுகளை உருவாக்க முடியாது? 'மனம் இருந்தால் மார்க்கம் உண்டு' என்ற பழமொழிக்கேற்ப நாம் தற்காலச் சூழலைக் கருத்தில் கொண்டு செயல்பட்டால், நிச்சயம் இந்தப் பூமியில் பெரும் அளவில் மரங்களை நட்டு வளர்க்க முடியும்.

வனங்களின் இன்றைய நிலை என்ன என்று நாம் சற்றே எண்ணிப் பார்க்கவேண்டும். உலகில் உள்ள வனங்கள் நிறைந்த நாடுகளில் இந்தியா பத்தாவது இடத்தைப் பெற்றுள்ளது. இந்தியாவின் பரப்பளவில் 23.4% காடுகள் மற்றும் மரங்கள் நிறைந்துள்ளன. தமிழ் நாட்டைப் பொறுத்தவரையில் இயற்கையான காடுகளின் பரப்பளவு 17.59%. தமிழ் நாடு வனத் துறையின் மூலம் செயல்படுத்தப்பட்டு வரும் பல்வேறு காடு வளர்ப்புத் திட்டங்கள் மூலம் காடு மற்றும் மரங்களின் பரப்பளவு தற்போது 23.9% அளவிற்கு உயர்ந்துள்ளது.



ஜாதவ் பாயங் மொலாய் இந்திய ஜனாதிபதியிடம் விருது பெறுதல்

ஒரு நாட்டின் அடர்ந்த வனப்பகுதிகளே இயற்கை வளம் செறிந்த பகுதிகளாகவும், உயிரியப்பன்மை நிறைந்து விளங்கும் இடங்களாகவும் திகழ்கின்றன. அதனால்தான் உலக நாடுகளும், ஒரு நாடு வளமுடன் திகழ வேண்டுமெனில் அந்நாட்டின் பரப்பில் மூன்றில் ஒரு பங்கு காடுகளாக இருக்க வேண்டுமென முடிவு செய்துள்ளன. இந்தியாவில் மேற்குத் தொடர் மலைப்பகுதிகளும், வடகிழக்கு இமயமலைப் பகுதிகளும் வனங்கள் செழித்து வளர்ந்து உயிரியப் பன்மையின் உறைவிடங்களாகத் திகழ்கின்றன. இங்கிருந்துதான் வற்றாத ஜீவ நதிகள் உருவாகி இந்தியாவை வளம் கொழிக்கச் செய்கின்றன.

இந்திய தேசிய வனக் கொள்கையின்படி நம்நாட்டின் நிலப்பரப்பில் 33% காடுகளாக இருக்கவேண்டும். அப்படிக் காடுகள் நிறைந்து இருந்தால்தான் நம்நாடு வளம்பெறும்; இங்கே வாழும் உயிரினங்கள் நலம் பெறும்.

ஆகவே நாம் அனைவரும் ஒன்றிணைந்து நமது வீட்டைச் சுற்றிலும், சாலை ஓரங்களிலும், தரிசு நிலங்களிலும், வயல்வெளி ஓரங்களிலும், குளக்கரைகளிலும் மரக்கன்றுகளை நட்டு வளர்த்து வனங்களைப் பாதுகாத்திட நம் நாட்டில் வளம் நிறைய நம் வாழ்வில் நலம் நிறையும்.



வ.சுந்தரராஜ் இ.வ.ப

மேனாள் துணை
வனப்பாதுகாவலர்

MEDICAL CAMPS IN TRIBAL VILLAGES OF MAHARASHTRA STATE



Photo Courtesy: www.pinterest.com

The State of Maharashtra has about 5.3 million tribal people as per the 2011 Census. There are around **45 tribal communities** spread across various districts of the state. The **Korku** population is predominant in the Melghat forest areas of Amravati District, while **Andhra, Pradhan, Gond, and Kolam** tribes are found mainly in Yavatmal District.

The **Worli** and **Katkri** tribes inhabit Thane and Palghar Districts, whereas the **Raja Gonds, Gonds, and Madia** form the main tribal groups in Chandrapur District.

Historically, the **Gond Rajas** ruled the region, and the **fort at Chandrapur** was built by one of their kings. The forests were once under the control of the Gond rulers before being transferred to the Government of Maharashtra.

The **Forest Department** has been a major source of employment for tribal people—offering work opportunities in timber harvesting, plantations, nursery operations, and other forestry-related activities. During the **early 1970s**, **malaria** and **malnutrition** were twin challenges

faced by the tribal communities. Recognizing the need for healthcare support, I took the initiative to organize **medical camps** for the forest dwellers during my service in Maharashtra.

Medical Camp 1 – Kamalapur, South Chanda Division

After my selection to the Indian Forest Service (IFS) in 1972, I resigned from my post as Assistant Professor of Botany at Government Arts College, Namakkal. Our IFS batch consisted of only ten officers, with combined training sessions alongside the State Forest Service batch.

Following completion of the AIFC Diploma Course at the Indian Forest College, Dehradun, and three months of training at the Lal Bahadur Shastri Academy, Mussoorie, I was allotted to Maharashtra State. My first posting was as Assistant Conservator of Forests (ACF) in East Melghat Division, Amravati District—then the top revenue-generating forest division. At that time, all timber harvesting was done by the Forest Department itself, with only a few coupes allotted to Forest Labour Cooperative Societies. Later, I was posted as Logging Officer in South Chanda Division at Allapalli, where I was promoted to Divisional Forest Officer (DFO).

A Christian Missionary Hospital near Allapalli, managed by doctors and nurses from Kerala, became a key partner in organizing our first medical camp at Kamalapur Range Headquarters. Forest Department personnel transported tribal villagers to the campsite in departmental lorries. The medical team treated

about 75 tribal men and women, mainly labourers engaged in felling and logging operations. Medicines and vitamin tablets worth Rs. 700 were distributed, funded by the Forest Department, while the hospital staff offered their services free of cost. In emergency cases, particularly involving pregnant women, I personally arranged for transport either in departmental lorries or the Allapalli Jeep to the Chandrapur Headquarters Hospital.

Medical Camp 2 – Kasa Tribal Village, Thane District

While serving as Regional Manager, Thane Region, under the Forest Development Corporation of Maharashtra (FDCM), we organized a medical camp at Kasa in 1993, utilizing funds from the World Food Programme (WFP). Doctors and nurses from Dahanu Division actively participated. Medicines, nutrition tablets, and food packets were distributed to the **Worli and Katkari** tribal communities. The event received an enthusiastic response from the villagers and was widely appreciated.

Medical Camp 3 – Tipeshwar, Pandharkawda Division, Yavatmal District

During my tenure as Chief Wildlife Warden (CWW), Maharashtra, I collaborated with a Christian Missionary Hospital at Umri village, which also ran a Nursing School offering diploma course. A young couple from Kanyakumari joined the institution in 2002, making it the only nursing school in Yavatmal District. The principal, who held a Ph.D. in Nursing, was eager to conduct a medical camp at Tipeshwar Forest Adivasi Village, located within the Tipeshwar Wildlife Sanctuary.

I directed Forest Department staff to assist the team, especially in transporting tribal villagers to the Forest Rest House (FRH) where the camp was organized. The medical team provided general treatments, medicines, and vitamin supplements. Unfortunately, the Principal performed hydrocele operations on a few villagers, which led to a Public Interest Litigation (PIL) since he was not a qualified surgeon. Despite this setback, the Nursing School's contribution was commendable—every student later secured a government job, with valuable support from both the Forest Department and the Missionary staff.

Medical Camp 4 – Semadoh, Melghat Tiger Project Area, Amravati District

As Chief Wildlife Warden, I was approached by Prof. Jabaraj from Christian Medical College (CMC), Vellore, who, along with six postgraduate medical students in orthopaedics, was conducting a study on malnutrition among pregnant tribal women (both prenatal and postnatal) in the Melghat region. The team visited the Government Hospital at Chikhaldara, interacted with expectant and new mothers, and collected data on healthcare access and welfare benefits under government schemes.

With their cooperation, I organized a medical camp at Semadoh Tribal Village, using rooms in the PWD Rest House for accommodation. About 305 tribal people attended the camp and received free medical treatment and nutrition tablets. Three patients were referred for surgery at the Missionary Hospital in Paratwada. This camp was one of the most successful, owing to the participation of qualified doctors and postgraduate trainees, and strong logistical support from the Forest Department.

Melghat, situated around Chikhaldara, is the highest point in Eastern Maharashtra and home to the **Korku tribe**. Later, while on deputation to the State Forest Service College, in Coimbatore, I took the first batch of SFS officer trainees to this picturesque Melghat forests and camped in Semadoh Tribal Village.

Post-Retirement Service – Medical Camps in Tamil Nadu

After my retirement, I continued organizing free medical camps in my native village, Peruvalur, a remote non-tribal area in Tamil Nadu. With the cooperation of government departments and voluntary doctors, I successfully conducted five medical camps for the local community. One of the most impactful initiatives was a **VASAN Eye Care Camp** held at **Peruvalur Village**, organized from Villupuram Headquarters. Doctors and technicians screened all **students from Classes I to VIII**, identifying those with vision problems and referring them to the hospital for further treatment.

The event was made possible through the support of the Village Panchayat President, the Headmaster Mr. Palani, and my friend Mr. G. Sampath (Retd. Headmaster).

My request.

My lifelong association with forest and tribal communities has been both professionally and personally rewarding. The experience of organizing medical and welfare camps in remote areas reaffirmed my belief that field officers can

make a tangible difference in the lives of underprivileged people through sustained efforts in healthcare, education, and environmental welfare. These humble initiatives, carried out with the cooperation of dedicated officers, medical professionals, and missionaries, continue to remind me that **true service lies beyond duty—in compassion and community upliftment.**

It is my request that young officers will do their utmost for the welfare of such under privileged forest tribes



K. Subramanian, IFS., (R)

Former Principal Chief Conservator
of Forests, Maharashtra

TASPEF Annual General body Meeting 14.09.2025

Felicitating outgoing and incoming office bearers

